Industrial AI (IAI) Final Project 2026

Project 1 – Semiconductor Etching Tool Health Assessment

1. Background

At a semiconductor manufacturing facility (FAB), they want to develop a health assessment system for their set of LAM 9600 TCP metal etchers. They want to detect anomalous or degraded behavior of the etching process and thus provide a reduction in downtime and higher yield for the factory. They currently have several sensors that are used to monitor the metal etcher including pressure, gas flow rates, and power, among other signals. The etcher exhibits considerable normal variation, and data was collected from 3 metal etchers (labeled experiment 1, experiment 2, experiment 3). These etchers are the same make and model, but do have some normal variation due to manufacturing variation, usage, etc.

They currently have collected baseline data from the 3 metal etchers. They would like to develop a health assessment model for each metal etcher. They want to use the multiple signals to provide an overall health indication that can detect if something is going wrong in the etching tool. This would include providing a health indicator value along with a threshold for detecting that the system is degraded and beyond what is considered normal. Although not required, they would also like to see which signal is contributing to the anomalous condition if a problem is detected. A contribution visual graph would help for determining the root-cause of the problem and speed up the time needed to do maintenance.

2. Problem Statement

Develop an automated health monitoring system for detecting anomalous conditions for each of the three metal etchers. Baseline data is provided from all three etchers. The health assessment method should provide a health indicator and also a threshold for determining whether the system is normal or degraded (not within the normal boundary).

3. Collected Trace Data

A set of 21 controller parameters are collected from the metal etcher, with 19 of the parameters being sensor signals and the first two parameters is the time and process step number respectively. The sampling rate is approximately 1Hz and an example trace signal from one of the 19 sensors is provided in Figure 1. Notice that in this study, there are 6 processes steps, but only the key steps (4 and 5 are included in the data collection routine. In this example trace signal, the signal is plotted for the two steps. It is common practice to extract summary statistics from each step from each of the trace signals. A list of the 21 parameters collected from the etcher is provided in Table 1.

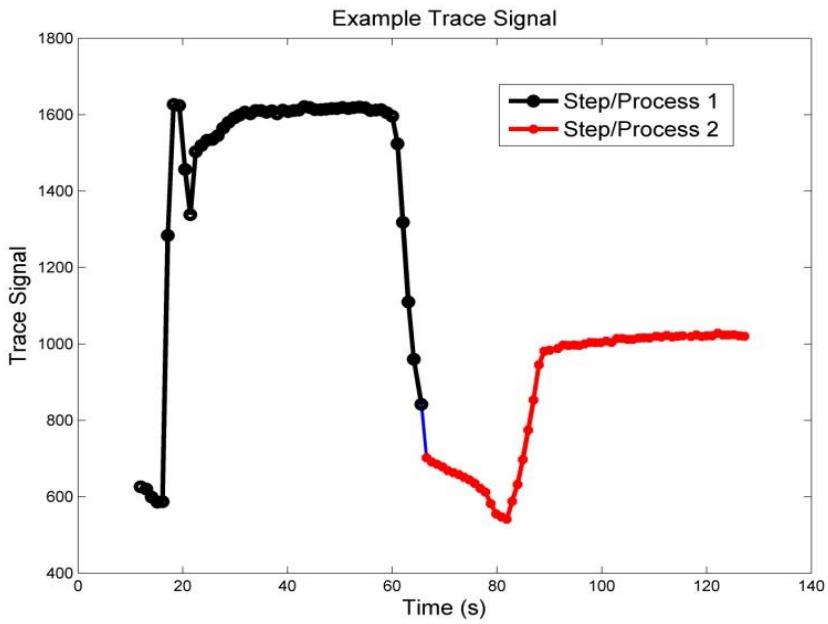


Figure 1: Example Trace Signal

Table 1: List of Variables Collected from Etcher

Variable #	Variable Name
1	Time
2	Step_Number
3	BCl3_Flow
4	Cl2_Flow
5	RF_Btm_Pwr
6	RF_Btm_Rfl_Pwr
7	Endpt_A
8	He_Press
9	Pressure
10	RF_Tuner
11	RF_Load
12	RF_Phase_Err
13	RF_Pwr
14	RF_Impedance
15	TCP_Tuner
16	TCP_Phase_Err
17	TCP_Impedance
18	TCP_Top_Pwr
19	TCP_Rfl_Pwr
20	TCP_Load
21	Vat_Valve

4. Provided Data Set

a) Training Data Set

There are 3 (.mat) files in the training data set folder. Each file contains the baseline data from each metal etcher. The structure (Baseline1Run) contains the 21 parameters from each baseline run.

Each run represents the processing of one wafer. There are 25 runs from each baseline data set. The structure (Baseline1ExpLabel) just indicates which etcher the data comes from, it is "0" for Baseline1 Data set, "1" for Baseline 2 Data Set, and "2" for Baseline 3 Data Set. It is just a book-keeping variable. The structure "VarName" contains the name of the 21 trace parameters (Table 1).

b) Testing Data Set

The testing data files are configured in the same way, but the health condition of each run is unknown. The objective is to determine for each run whether the etching process is normal or abnormal (degraded/anomalous). There are 18 runs for the first testing data set, 15 runs for the second testing data set, and also 15 runs for the third testing data set. There are three testing data sets, since each data set corresponds to the different etcher (e.g. Baseline1 and Testing 1 are from the same etcher).

5. Suggested Algorithms

a) Feature Extraction

a. Summary Statistics for Each Signal and Step

b) Feature Reduction

a. Principal Component Analysis (PCA)

c) Health Assessment

a. Principal Component Analysis Monitoring Method (T2 / SPE)

b. Self-Organizing Map MQE Method

6. Submission of Test Results

The group will submit the results for the test files (in excel or word or text file) stating the health condition of the test runs for each test data set (3 test data sets, 18, 15, 15 runs respectively). Indicate a 0 for healthy and 1 for not healthy/abnormal for each run.